INCH-POUND

MIL-D-20505D (AR)
AMENDMENT 2
16 August 1995
SUPERSEDING
AMENDMENT 1
10 January 1986

MILITARY SPECIFICATION

DETONATION, SIMULATOR, EXPLOSIVE, M80,
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-D-20205D (AR), dated 15 May 1984, and is approved for use by the U.S. Army Armament, Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 8

4.4.2.1, Minor 201: Delete "inside diameter" and replace with "wall thickness".

PAGE 9

4.4.2.2, Minor 201: Delete "outside diameter" and replace with "wall thickness".

PAGE 10

* 4.4.2.3: Body Assembly, Minor 201, delete "Crimp length" and substitute "Crimp length (when applicable)".

* Add Minor 204: "204 Epoxy missing or improperly applied (when applicable) 0.65% 3.2 Visual".

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* 4.4.2.4: Under Title, delete "Assembly" and substitute "Detonation, Simulator, Explosive, M80".

* Major 101: Delete "in assembly" and substitute "in simulator".

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* 4.4.3.1.2: Complete assembly. Delete in its entirety.

PAGE 16

4.5.1.1: Delete entire paragraph and replace with the following:

"4.5.1.1 Moisture content of charge composition. An accurately weighed portion of (3) grams (g.) of the charge composition shall be transferred to a tared dish. The moisture dish and contents shall be placed in a drying oven at 70 ± 2°C for 4 hours, cooled in a desiccator and weighed. The percentage of moisture in the sample shall be calculated as follows:

$$\text{Percentage of Moisture} = \frac{100A}{W}$$

Where:

A = loss in weight of moisture dish and contents.

W = weight of sample, g."

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* 4.5.1.3: Moisture content of complete assembly. Delete in its entirety.

4.5.2: Delete entire paragraph and replace with the following:

"4.5.2 Charge composition analysis.

4.5.2.1 Procedure. Transfer an accurately weighed portion of approximately 2 grams of the flash powder that was dried for the moisture determination to a tared crucible. Wash the sample between 10 and 15 times with 5 milliliter portions of cold carbon disulfide and rinse with ether. Continue washing the sample with fifteen, 20 milliliters portions of cold distilled water, allowing each wash of the water to remain in contact with the sample approximately 1 minute before applying suction. Aspirate the crucible and contents thoroughly, and dry in an oven maintained at a temperature of 100 plus 5 degrees C. The crucible and contents shall be cooled in a desiccator and weighed. To the contents remaining in the crucible, add several portions of hot dilute hydrochloric acid until the reaction subsides. After all reaction has ceased, wash the residue with several portions of hot water. Dry the crucible and contents in the 100 plus 5 degree C oven and then cool in a desiccator and weigh. Calculate the percent of aluminum as follows:

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Percent aluminum = (A-B) 100 divided by W

Where:

A = weight of crucible and contents after the carbon disulfide and water extraction.

B = weight of crucible and contents after the acid extraction.

C = weight of the sample.

Determine the percentage of black powder by subtracting the percentage of aluminum from 100."

4.4.2.4, Add paragraph:

"4.5.2.4 Alternative method. An alternate analysis by direct atomic absorption spectrometer measurement per MIL-STD-286 method T316.1 may be used. The inspection equipment and procedures shall be submitted for approval. (See 6.3)"

Note: The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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